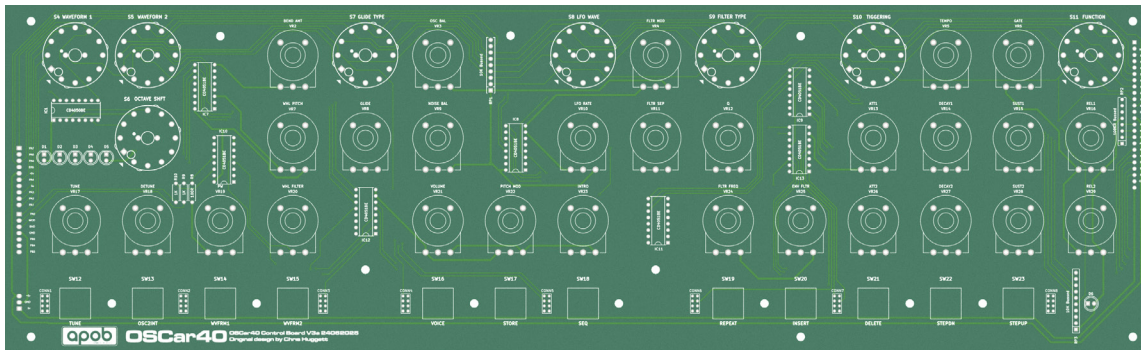


2 ASSEMBLY HINTS - PCBS

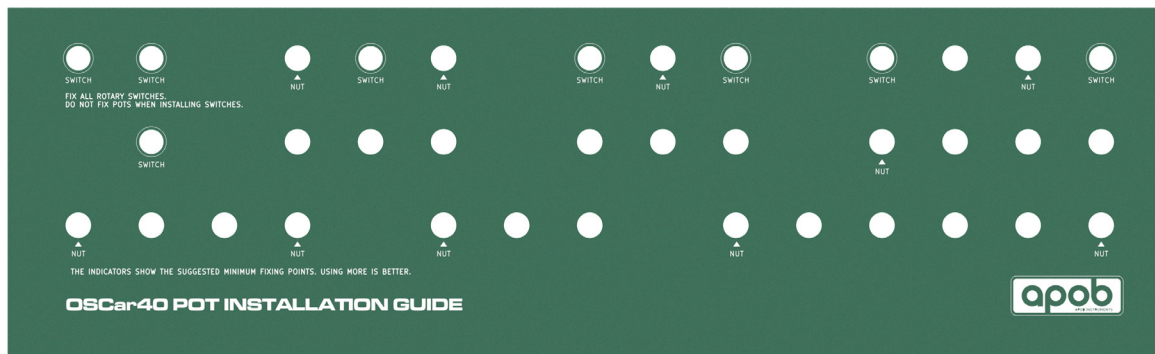


Main control board (CNTRL)

Suggested install order:

1. Resistors
2. 8 pin male connectors (for tact switch riser boards)
3. IC sockets
4. SIP resistor arrays
5. Potentiometers (use the pot guide board)
6. Rotary switches (use the pot guide board)
7. LED's (with spacers)
8. Connectors mounted on *underside*.

I've used sockets for all IC's on my builds (25+) as it makes it easier to troubleshoot problems or rectify mistakes. This is more important on the processor (PROC) board than the CNTRL board. If you're confident and experienced you could skip the sockets.



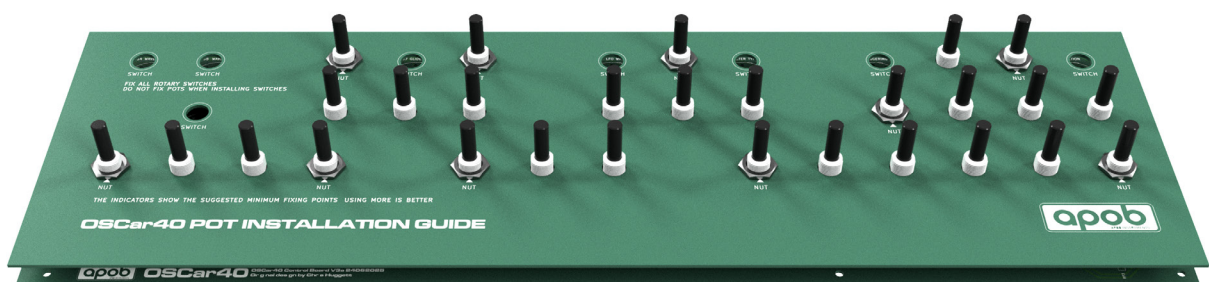
Installing the potentiometers

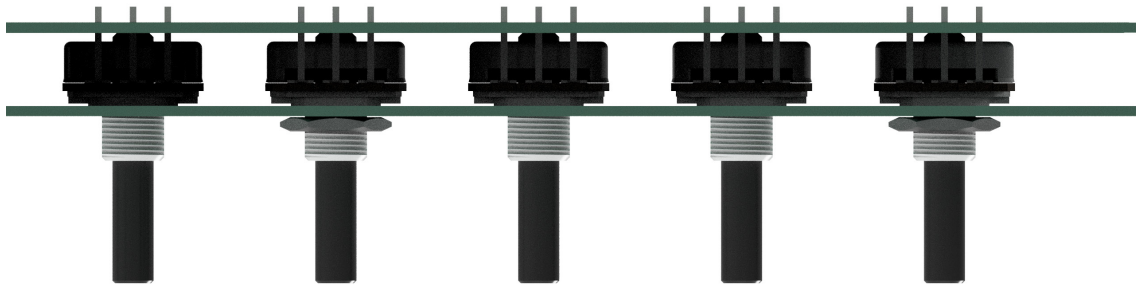
The rotary switch bushings sit higher than those of the potentiometers so the potentiometers have to be installed first, followed by the rotary switches.

The pot guide ensures that the potentiometers are accurately positioned so that their shafts are centered with the openings in the enclosure when the CNTRL PCB is installed. This precision is important because the bottom of each knob sits below the top panel, so good alignment is essential for achieving a even result.

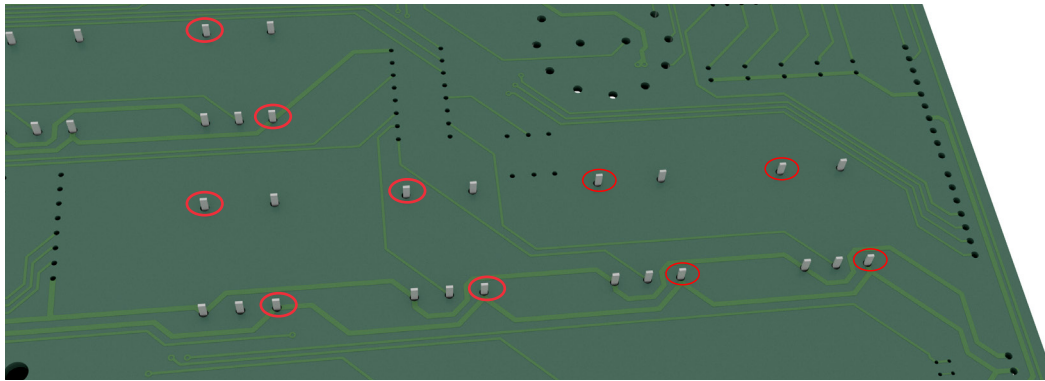
When installing the potentiometers, lay the PCB on a thickly folded towel. This setup allows the pot pins to pass completely through the PCB holes, ensuring the pots fit snugly and sit evenly on the PCB. With all the pots installed:

- Place the pot guide PCB over the top of the installed pots.
- Ensure each pot has pushed through until it sits flush with the bushing.
- Use the supplied pot nuts to secure the pot guide PCB in place.
- Refer to the pot guide board for the recommended securing positions. At a minimum, secure the pots as indicated.
- For added stability, you can secure additional pots beyond the minimum as desired.





Now flip the joined CNTRL and guide PCBs over so that the bottom of the CNTRL PCB is facing upward. Go around the board pushing down on the bottom of the pots to make sure they're all fitting tightly and evenly. Check at eye-level that the two boards are evenly spaced along the length of each side.

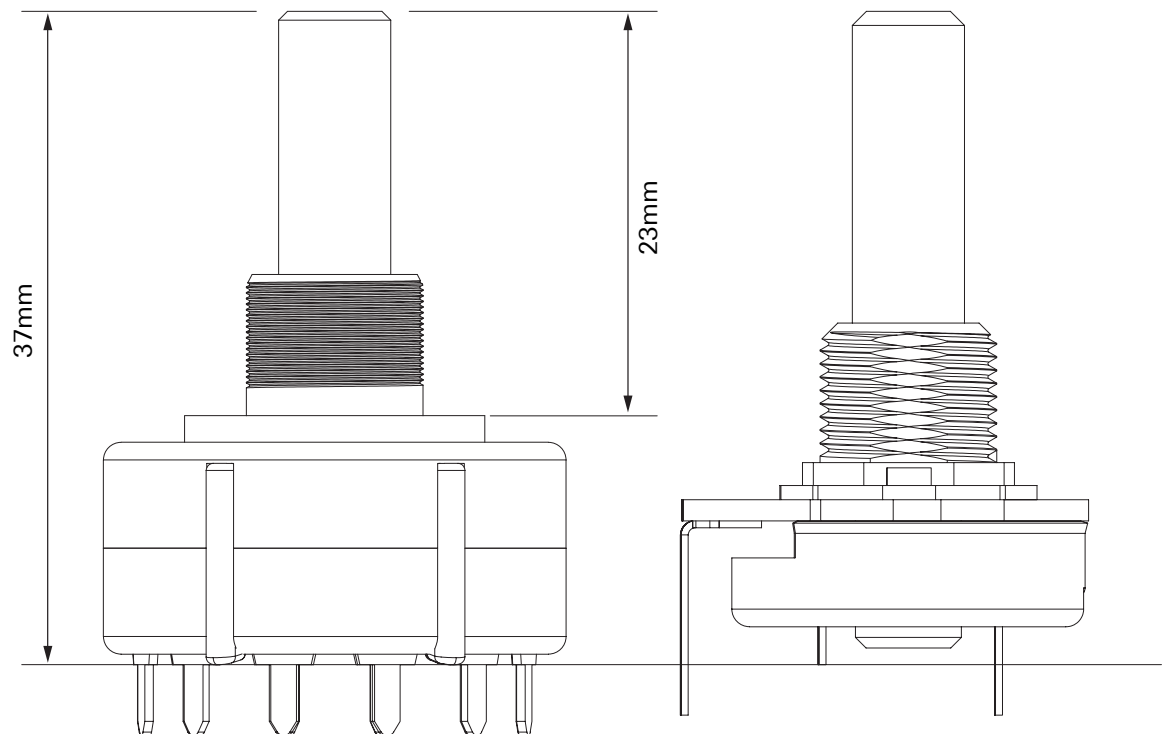


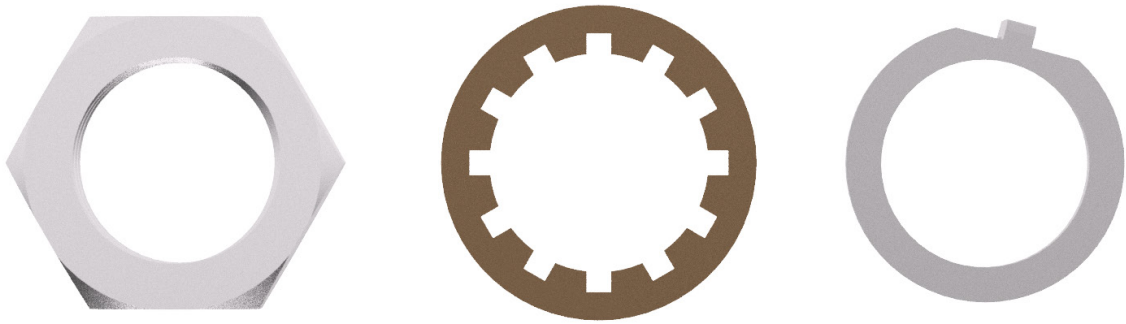
- Solder only the diagonally opposite pads for each potentiometer.
- Once these pads are soldered, remove the pot guide.
- If a pot isn't aligned with others in the row or column, heat one of the soldered pads and adjust the pot's position to bring it in line.
- Careful use of the pot guide and proper seating usually means little or no adjustment is needed.
- This method helps keep all pots straight and consistent on your PCB.
- When you're happy with how the pots are sitting you can solder the remainder of the pot pads.

Rotary switches - Trimming the shafts

You will need to match the rotary switch shaft height to the potentiometers:

- The potentiometer shafts extend 37mm above the PCB. Your goal is to make the rotary switch shaft tops level with these.
- Measure 23mm from the base of each rotary switch bushing (where the threaded part meets the switch body) and mark this point on the shaft.
- Cut the shaft at the 23mm mark.
- After cutting, sand the end of each shaft to create a small chamfer. This smooths the edge and makes it easier to fit the knobs onto the shafts.





Rotary switches - Adjusting the number of positions

The rotary switches come with a nut, a locking washer and an adjustment ring to set the the number of postions. By default the switch is set for 12 positions, this needs to be set to 6.

- Remove the nut from the rotary switch.
- Slide off the locking washer from the shaft.
- Remove the adjustment ring.



Figure 1

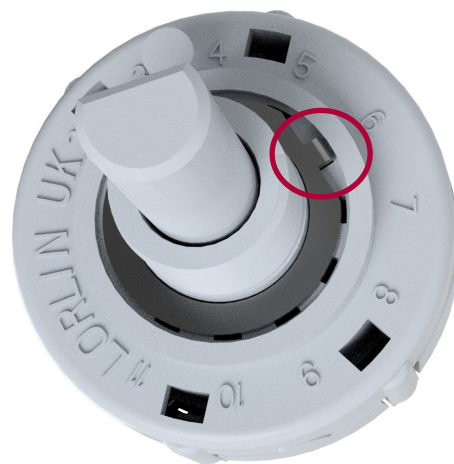
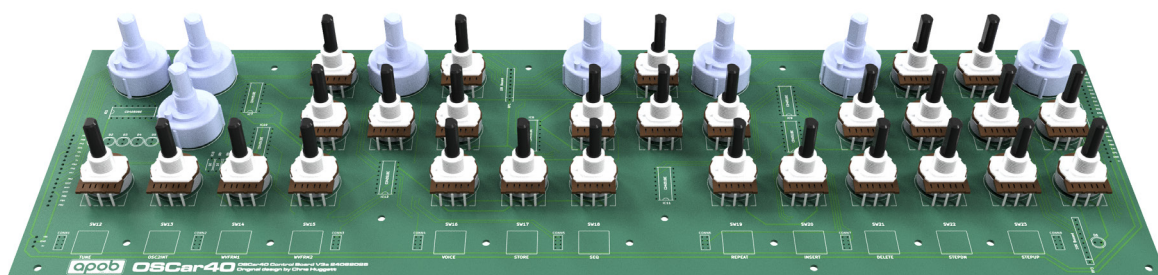


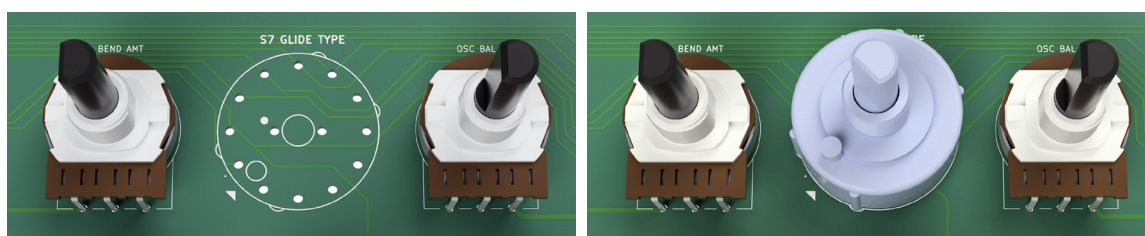
Figure 2

- Locate the hole for position 6 (as in ***Figure 1*** above).
- Replace the adjustment ring with the lug inserted into the hole for position 6. (as in ***Figure 2*** above).
- Replace the locking washer.
- Replace and hand tighten the nut to keep everything in place.



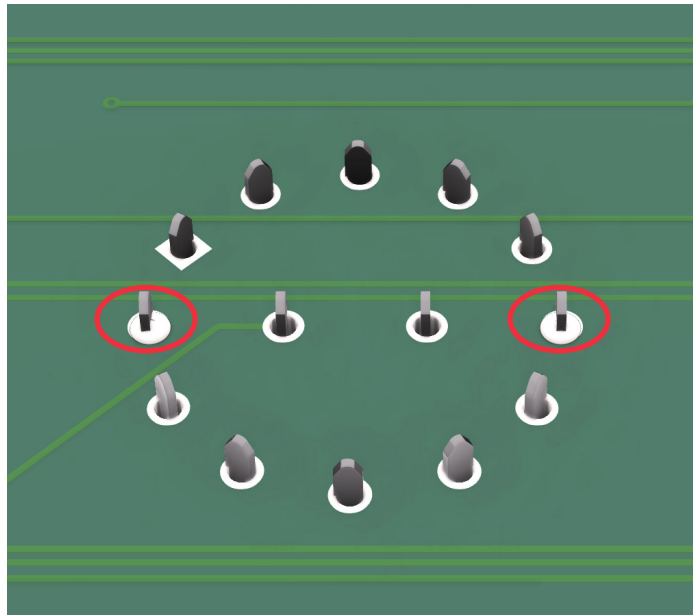
Installing the rotary switches

When installing the rotary switches, make sure to orient them correctly. Pay attention to the location of the locating lug on each switch, and ensure it matches the position as indicated in the reference image below.



- Once all the rotary switches are installed, ensure each one sits flat on the board.
- Carefully remove the nuts from each rotary switch, taking care not to dislodge the adjustment rings.
- Gently place the pot guide over the board, making sure not to knock or move the rotary switches out of place.
- Replace the nuts onto each rotary switch and hand-tighten them to secure the switches to the pot guide.
- Place upside down and press firmly around the pins of each switch to make sure they are sitting flush with the CNTRL PCB. (Be careful when turning over not to dislodge the rotary switches).





- Solder only the opposite pads for each rotary switch.
- Once these pads are soldered, remove the pot guide.
- If a switch isn't aligned with others in the row or column, heat one of the soldered pads and adjust the switches position to bring it in line.